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SCHOOL : INFORMATION SCIENCE AND TECHNOLOGY

PROGRAM : SOFTWARE ENGINEERING

MODULE : OBJECT ORIENTED PROGRAMMING

ASSIGNMENT : PRACTICAL

Task 1: Account.java

```
package Bank;

public class Accounts {

    public Accounts() {

        public abstract class Account {
            protected String accountNumber;
            protected double balance;

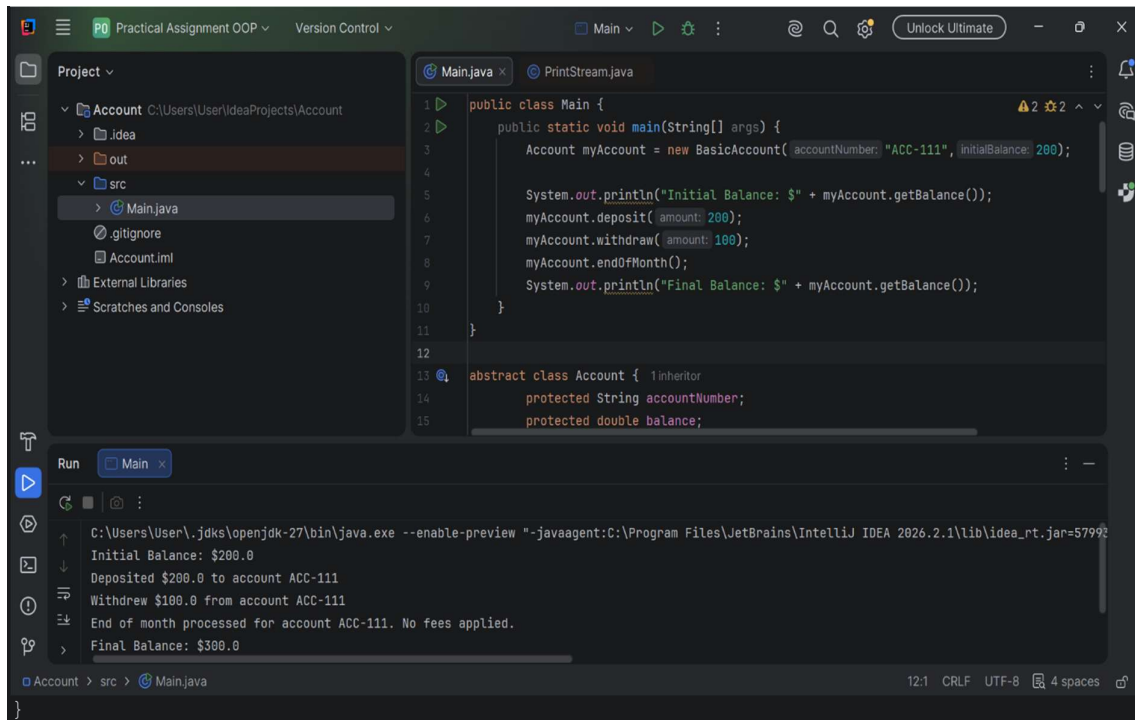
            public Account(String accountNumber, double initialBalance) {
                this.accountNumber = accountNumber;
                this.balance = initialBalance;
            }

            public void deposit(double amount) {
                if (amount > 0) {
                    balance += amount;
                    System.out.println("Deposited $" + amount + " to account "
+ accountNumber);
                } else {
                    System.out.println("Deposit failed: Amount must be
positive.");
                }
            }

            public double getBalance() {
                return balance;
            }

            public abstract void withdraw(double amount);

            public abstract void endOfMonth();
        }
    }
}
```



Task 2: SavingsAccount.java

```

package Bank;

public class SavingsAccount {

    public SavingsAccount() {

        public class SavingsAccount extends Account {
            private double minimumBalance;
            private double interestRate;

            public SavingsAccount(String accountNumber, double
initialBalance, double minimumBalance, double interestRate) {
                super(accountNumber, initialBalance);
                this.minimumBalance = minimumBalance;
                this.interestRate = interestRate;
            }

            @Override
            public void withdraw(double amount) {
                if (balance - amount < minimumBalance) {
                    System.out.println("Withdrawal of $" + amount + " rejected.
Account " + accountNumber + " must maintain a minimum balance of $" +
minimumBalance);
                } else {
                    balance -= amount;
                    System.out.println("Withdrew $" + amount + " from account "
+ accountNumber);
                }
            }
        }
    }
}

```

```

    }

    @Override
    public void endOfMonth() {
        double interest = balance * interestRate;
        balance += interest;
        System.out.println("End of month processing: Added $" +
interest + " interest to account " + accountNumber);
    }
}
}
}

```

The screenshot shows the IntelliJ IDEA IDE with a project named 'Practical Assignment OOP'. The 'Project' view on the left shows the file structure: 'Account' (C:\Users\User\IdeaProjects\Account) containing 'idea', 'out', 'src', 'Main.java', '.gitignore', and 'Account.iml'. The 'Main.java' file is open in the editor, showing the following code:

```

1 public class Main {
2     public static void main(String[] args) {
3         SavingsAccount mySavings = new SavingsAccount("SA-001", 1000.0);
4
5         System.out.println("Initial Balance: $" + mySavings.getBalance());
6         System.out.println("-----");
7
8         mySavings.deposit(200.0);
9         mySavings.withdraw(500.0);
10        mySavings.withdraw(650.0);
11        mySavings.endOfMonth();
12    }
13 }

```

The 'Run' view at the bottom shows the execution output for the 'Main' class:

```

C:\Users\User\.jdk\openjdk-27\bin\java.exe --enable-preview --javaagent:C:\Program Files\JetBrains\IntelliJ IDEA 2026.2.1\lib\idea_rt.jar=5070
Initial Balance: $1000.0
-----
Deposited $200.0 to account SA-001
Withdraw $500.0 from account SA-001
Withdrawal of $650.0 rejected. Account SA-001 must maintain a minimum balance of $100.0
End of month processing: Added $35.0 interest to account SA-001
-----
Final Balance: $735.0

```

Task 3: CurrentAccount.java

```

package Bank;

public class CurrentAccount {

    public CurrentAccount() {
        public class CurrentAccount extends Account {
            private double overdraftLimit;
            private double maintenanceFee;

```

```

        public CurrentAccount(String accountNumber, double
initialBalance, double overdraftLimit, double maintenanceFee) {
            super(accountNumber, initialBalance);
            this.overdraftLimit = overdraftLimit;
            this.maintenanceFee = maintenanceFee;
        }

        @Override
        public void withdraw(double amount) {
            if (balance - amount < -overdraftLimit) {
                System.out.println("Withdrawal of $" + amount + " rejected.
Overdraft limit of $" + overdraftLimit + " exceeded for account " +
accountNumber);
            } else {
                balance -= amount;
                System.out.println("Withdrew $" + amount + " from account "
+ accountNumber);
            }
        }

        @Override
        public void endOfMonth() {
            balance -= maintenanceFee;
            System.out.println("End of month processing: Deducted $" +
maintenanceFee + " maintenance fee from account " + accountNumber);
        }
    }
}

```

The screenshot displays the IntelliJ IDEA IDE with the following components:

- Project View:** Shows the project structure with folders for `idea`, `out`, `src`, and files for `Main.java`, `Account.iml`, `gitignore`, `External Libraries`, and `Scratches and Consoles`.
- Main.java:** Contains the following code:


```

1 public class Main {
2     public static void main(String[] args) {
3         CurrentAccount myCurrent = new CurrentAccount("CA-999", 500.0);
4
5         System.out.println("Initial Balance: $" + myCurrent.getBalance());
6         System.out.println("-----");
7
8         // Test inherited deposit method
9         myCurrent.deposit(100.0);
10
11         // Test normal withdrawal
      
```
- Run Output:** Shows the execution results:


```

C:\Users\User\.jdk\openjdk-27\bin\java.exe --enable-preview "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA 2026.2.1\lib\idea_rt.jar=63288
Initial Balance: $500.0
-----
Deposited $100.0 to account CA-999
Withdrew $400.0 from account CA-999
Withdrew $300.0 from account CA-999
Withdrawal of $200.0 rejected. Overdraft limit of $200.0 exceeded for account CA-999
End of month processing: Deducted $15.0 maintenance fee from account CA-999
-----
Final Balance: $-115.0
      
```

Task 4: BankDemo.java

```
package Bank;

import java.util.ArrayList;
import java.util.List;

public class BankDemo {
    public static void main(String[] args) {

        List<Account> accounts = new ArrayList<>();

        accounts.add(new SavingsAccount("SA-101", 1000.0, 100.0, 0.05));
        accounts.add(new CurrentAccount("CA-201", 500.0, 200.0, 15.0));

        System.out.println("--- Polymorphic Loop Demonstration ---");

        for (Account acc : accounts) {
            System.out.println("Processing Account: " +
acc.accountNumber);
            acc.withdraw(50.0);
            acc.endOfMonth();
            System.out.println("Current Balance: $" + acc.getBalance());
            System.out.println("-----");
        }

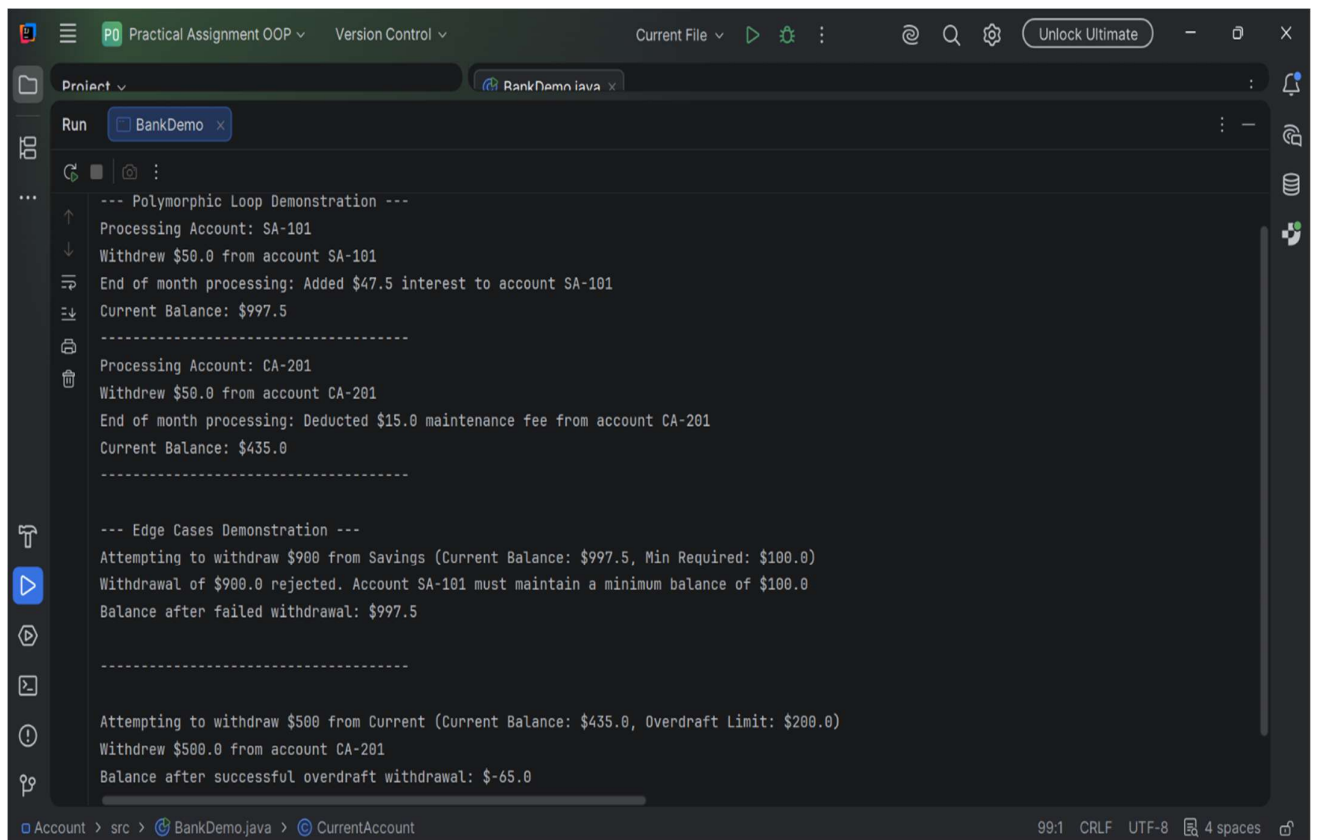
        System.out.println("\n--- Edge Cases Demonstration ---");
        Account savings = accounts.get(0);
        Account current = accounts.get(1);

        System.out.println("Attempting to withdraw $900 from Savings
(Current Balance: $" + savings.getBalance() + ", Min Required: $100.0)");
        savings.withdraw(900.0);
        System.out.println("Balance after failed withdrawal: $" +
savings.getBalance());

        System.out.println("\n-----\n");

        System.out.println("Attempting to withdraw $500 from Current
(Current Balance: $" + current.getBalance() + ", Overdraft Limit:
$200.0)");
        current.withdraw(500.0);
        System.out.println("Balance after successful overdraft
withdrawal: $" + current.getBalance());
    }
}
```

Task 5: Sample outputs showing the polymorphic loop and both edge cases



```
--- Polymorphic Loop Demonstration ---
Processing Account: SA-101
Withdraw $50.0 from account SA-101
End of month processing: Added $47.5 interest to account SA-101
Current Balance: $997.5
-----
Processing Account: CA-201
Withdraw $50.0 from account CA-201
End of month processing: Deducted $15.0 maintenance fee from account CA-201
Current Balance: $435.0
-----

--- Edge Cases Demonstration ---
Attempting to withdraw $900 from Savings (Current Balance: $997.5, Min Required: $100.0)
Withdrawal of $900.0 rejected. Account SA-101 must maintain a minimum balance of $100.0
Balance after failed withdrawal: $997.5
-----

Attempting to withdraw $500 from Current (Current Balance: $435.0, Overdraft Limit: $200.0)
Withdraw $500.0 from account CA-201
Balance after successful overdraft withdrawal: $-65.0
```

Account > src > BankDemo.java > CurrentAccount 99:1 CRLF UTF-8 4 spaces